

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 04

Deploy Multi-Region Security Alert Topics with AWS CloudFormation

AWS CloudFormation | Amazon SNS | Infrastructure as Code

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.1
RELEASE DATE	July 8, 2026
STATUS	Complete

Executive Summary

Deployed AWS CloudFormation stacks in **us-west-2 (Oregon)** and **us-east-1 (N. Virginia)** to create identically named Amazon SNS topics called **SecurityAlerts**. The project introduced Infrastructure as Code (IaC), demonstrated template portability across regions, and established reusable notification infrastructure for future cloud security alerts.

Business Problem

Future security controls need a consistent notification destination for alerts. Creating those resources manually in multiple regions increases the risk of configuration drift and naming inconsistencies. A repeatable Infrastructure as Code deployment provides a deterministic way to create the same security notification resource wherever it is needed.

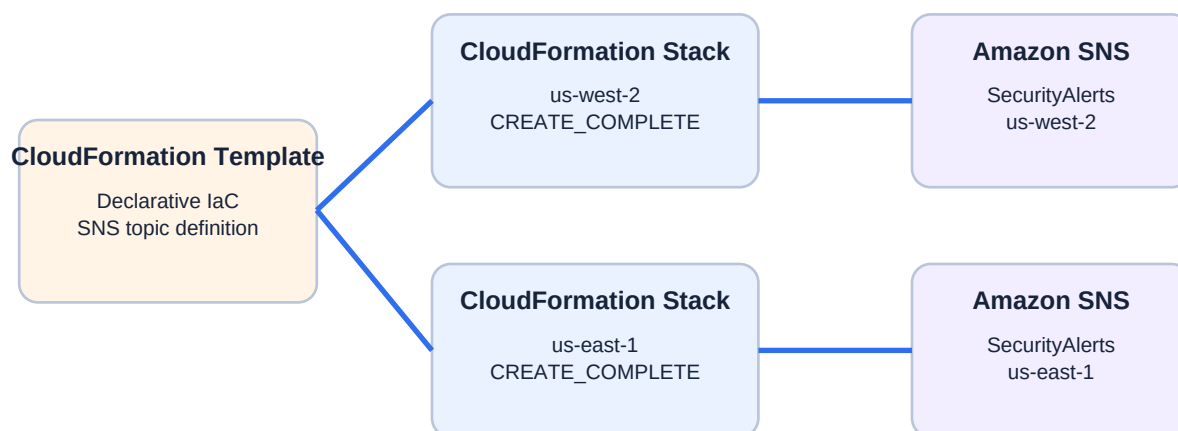
Objectives

- Use AWS CloudFormation to deploy infrastructure from a declarative template.
- Create a SecurityAlerts Amazon SNS topic in us-west-2.
- Repeat the deployment in us-east-1 to demonstrate regional portability.
- Validate successful stack creation in both regions.
- Establish notification infrastructure for future security monitoring projects.

Environment

Cloud Provider	Amazon Web Services (AWS)
Infrastructure as Code Service	AWS CloudFormation
Deployed Resource	Amazon SNS topic named SecurityAlerts
Regions	us-west-2 (Oregon) and us-east-1 (N. Virginia)
Stack Status	CREATE_COMPLETE in both regions
Template Format	YAML syntax shown in the provided template evidence
Purpose	Reusable security alert notification foundation

Architecture



The same infrastructure definition was deployed in two AWS regions to create identically named SecurityAlerts SNS topics.

Figure 1 - Multi-region CloudFormation deployment architecture.

Implementation Summary

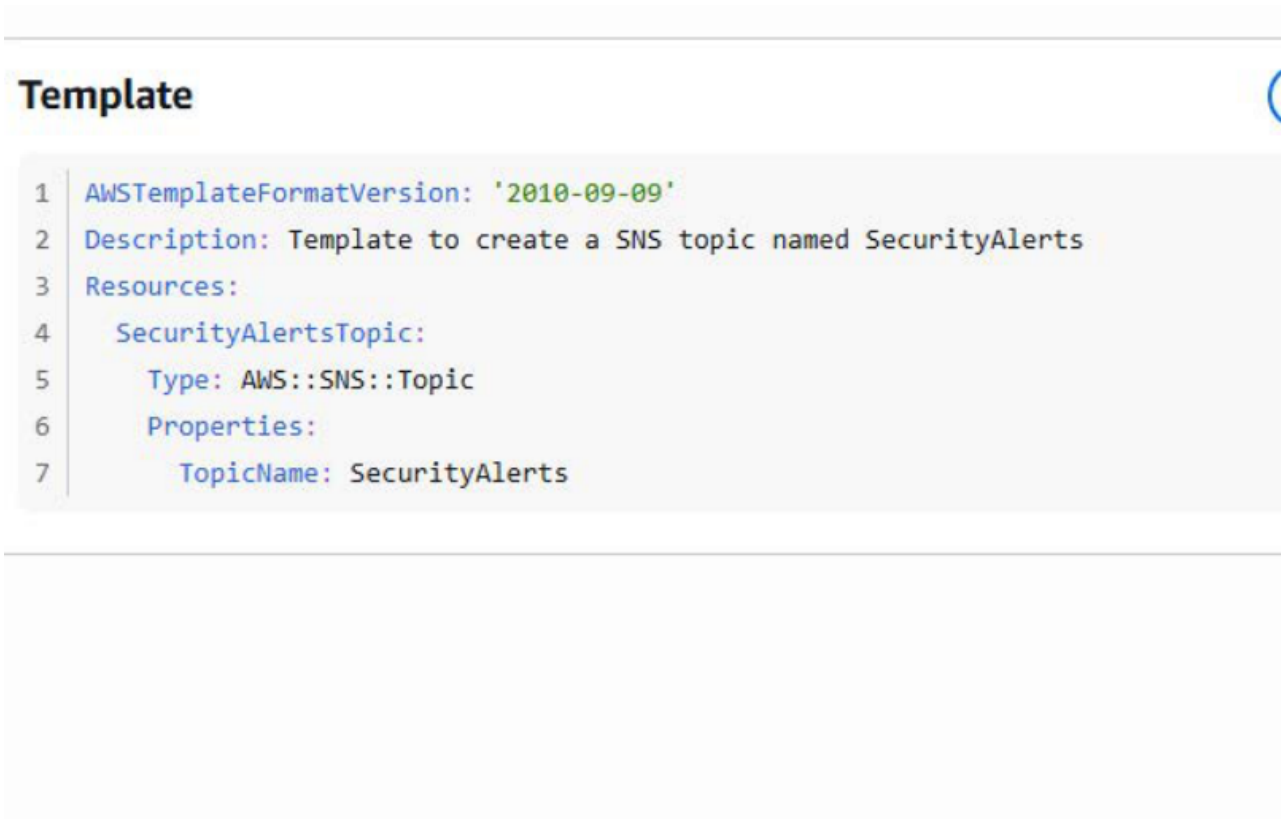
1. Reviewed the CloudFormation template before deployment to understand the resource that would be created.
2. Deployed the SecurityAlerts stack in us-west-2.
3. Validated that the stack reached CREATE_COMPLETE.
4. Repeated the same CloudFormation deployment in us-east-1.
5. Validated that the second stack also reached CREATE_COMPLETE.
6. Confirmed the template defines an AWS::SNS::Topic resource named SecurityAlerts.

DOCUMENTATION NOTE

The implementation note described the template as JSON, but the uploaded template screenshot is YAML-formatted. This document records the format shown by the evidence while recognizing that CloudFormation supports both JSON and YAML.

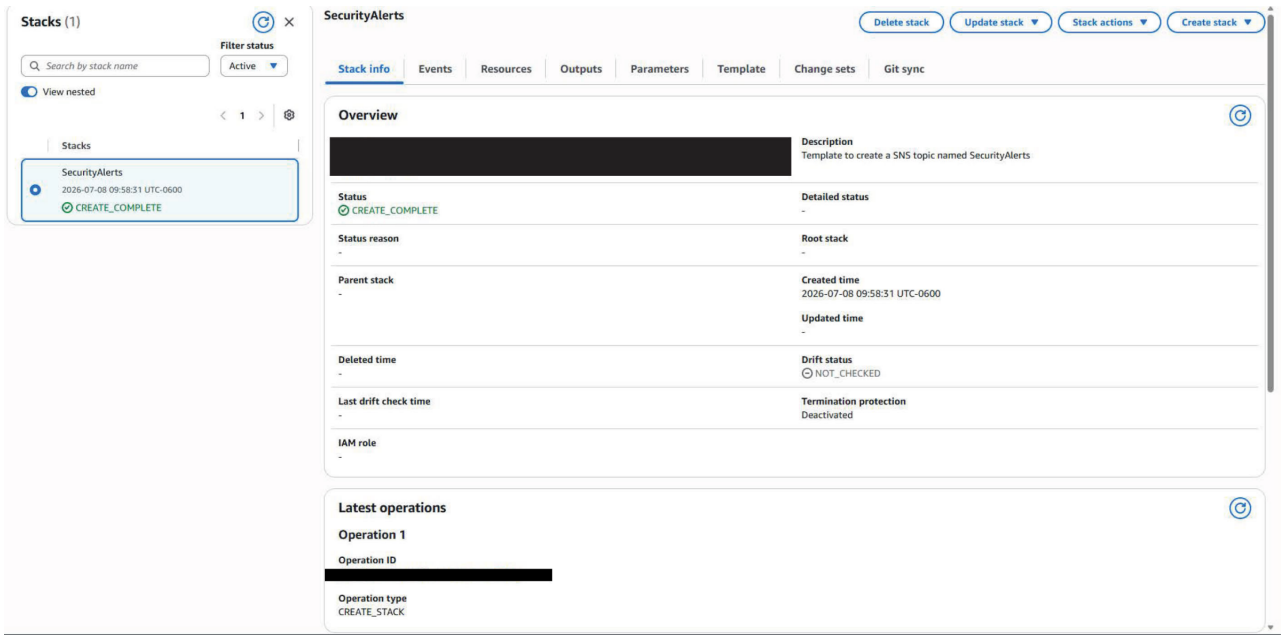
Evidence

Figure 2 - CloudFormation Template for the SecurityAlerts Topic



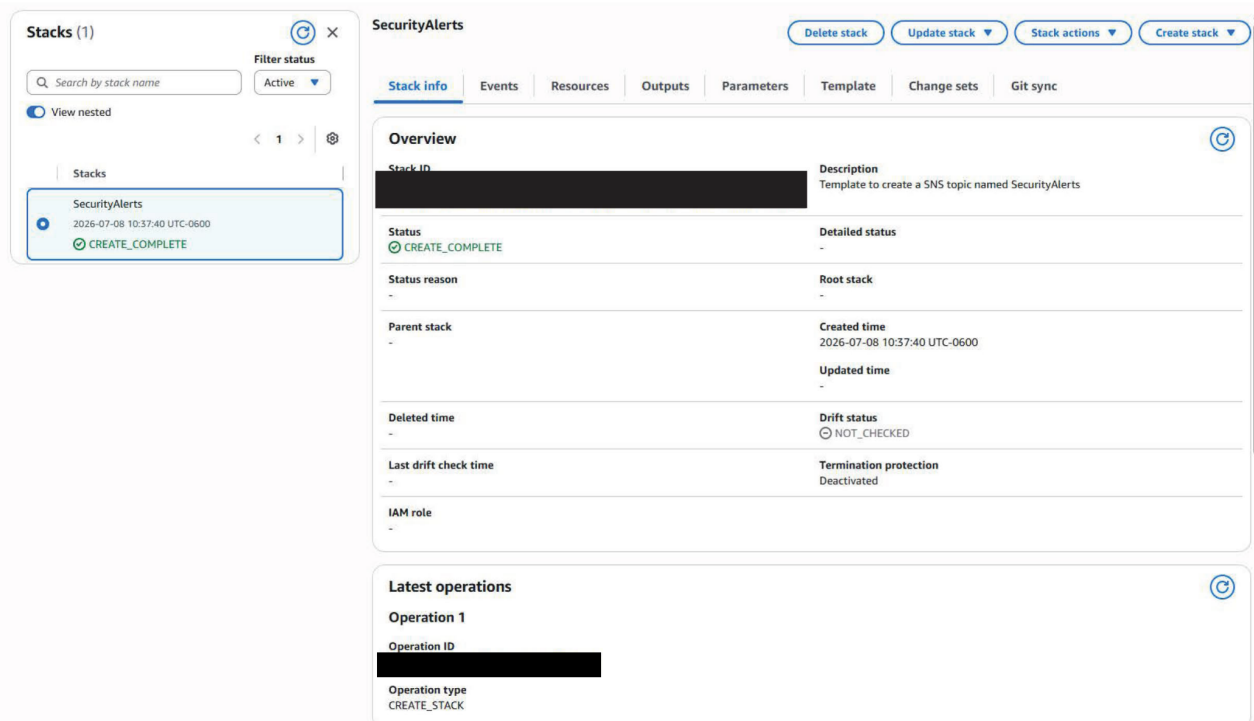
The template evidence defines one `AWS::SNS::Topic` resource with the `TopicName` property set to `SecurityAlerts`. The concise declarative definition makes the intended infrastructure easy to review before deployment.

Figure 3 - US West (Oregon) CloudFormation Stack



The replacement redacted CloudFormation screenshot shows the `SecurityAlerts` stack in a `CREATE_COMPLETE` state for the US West deployment. Sensitive stack and operation identifiers are fully redacted.

Figure 4 - US East (N. Virginia) CloudFormation Stack



The replacement redacted CloudFormation screenshot shows a second SecurityAlerts stack in a CREATE_COMPLETE state for the US East deployment. Sensitive stack and operation identifiers are fully redacted.

Security Analysis

Infrastructure as Code improves repeatability and gives administrators a declarative record of intended infrastructure. This can reduce configuration drift and make security reviews more consistent.

CloudFormation templates should be reviewed before deployment, especially when hosted or authored by a third party. A malicious or modified template can create privileged resources, expose data, or deploy unwanted workloads.

Deploying the same notification topic in two regions creates a consistent alerting foundation for services that operate regionally and for future controls that may need to publish security events from either region.

Lessons Learned

A CloudFormation stack is a managed collection of resources created from a template. The same template can be reused to build a consistent environment in different regions.

IaC is valuable not only for deployment speed but also for security. The template provides a reviewable baseline that can be version-controlled, compared, and reused during recovery or rebuild activities.

Regional services require regional planning. Creating SecurityAlerts in both us-west-2 and us-east-1 prepares the environment for later security alert integrations.

Production Improvements

- Store the CloudFormation template in a version-controlled repository and review changes through pull requests.
- Run template validation and linting before deployment.
- Use CloudFormation change sets to preview resource changes before updating an existing stack.
- Consider StackSets for repeatable multi-account or larger multi-region deployments.
- Apply an SNS topic policy and encryption settings appropriate to production security requirements.
- Add subscriptions and integrate the topics with monitoring and detection services in later projects.

Skills Demonstrated

AWS CloudFormation	Template review	Infrastructure as Code
Amazon SNS	Repeatable deployment	Multi-region provisioning
Stack validation	Security alert architecture	Declarative infrastructure

Resume Bullet

RESUME

Deployed AWS CloudFormation stacks in us-west-2 and us-east-1 to create standardized SecurityAlerts SNS topics, demonstrating Infrastructure as Code, repeatable multi-region provisioning, and a foundation for future security alerting.

STAR Interview Story

Situation: I needed a consistent notification resource in two AWS regions for later cloud security alerting projects.

Task: Create the same security alert topic in both regions without relying on separate manual configurations.

Action: I reviewed the CloudFormation template, deployed the SecurityAlerts stack in us-west-2, validated CREATE_COMPLETE, repeated the deployment in us-east-1, and confirmed the second stack completed successfully.

Result: I established repeatable multi-region SNS infrastructure and gained hands-on experience using CloudFormation as AWS's native Infrastructure as Code service.

Version History

Version	Deployment Date	Deployment Details
v1.0.0	July 8, 2026	Initial documented implementation. Deployed SecurityAlerts CloudFormation stacks in us-west-2 and us-east-1, validated CREATE_COMPLETE in both regions, and embedded the original project evidence.
v1.0.1	July 8, 2026	Replaced the US West and US East CloudFormation stack screenshots with newly provided redacted versions that fully obscure stack and operation identifiers. No infrastructure changes were made.

References

- Cloud Security Lab a Week (S.L.A.W.) - Timmy's First CloudFormation.
- User-provided replacement redacted CloudFormation stack screenshot for us-west-2.
- User-provided replacement redacted CloudFormation stack screenshot for us-east-1.
- User-provided CloudFormation template screenshot.